# CSE 400 – Lecture Scribe

School of Engineering and Applied Sciences, Ahmedabad University

**Lecture 10: Min-Cut Algorithms (Deterministic and Randomized)**

## 1. Min-Cut Problem

### 1.1 Why Use Min-Cut?

**Step-by-step reasoning (as per lecture):**

1. The min-cut algorithm is used to solve problems related to:
   - Network connectivity
   - Reliability
   - Optimization
2. In **network design**, min-cut:
   - Helps improve efficiency of communication
   - Optimizes network flow
   - Finds the minimum capacity cut
3. In **communication networks**, min-cut:
   - Helps understand vulnerability of networks to failures
   - Assists in building robust and fault-tolerant networks
4. In **VLSI design**, min-cut:
   - Helps partition circuits into smaller components
   - Reduces interconnectivity complexity

### 1.2 What Is Min-Cut?

**Definitions (step-by-step):**

1. A **cut-set** in a graph is:
   - A set of edges whose removal breaks the graph into two or more connected components
2. Given a graph
   $$G = (V, E)$$
   with $n$ vertices:
   - The **minimum cut (min-cut)** problem is to find a cut-set with **minimum cardinality**
3. Min-cut algorithms (e.g., Karger's algorithm):
   - Are random
   - Can be sensitive to the initial choice of edges
4. If critical edges are contracted early:
   - The algorithm may find a smaller cut

### 1.3 Edge Contraction (Core Operation)

**Step-by-step operation:**

1. The main operation is **edge contraction**
2. Contracting an edge $(u, v)$:
   - Merge vertices $u$ and $v$ into a single vertex
   - Eliminate all edges connecting $u$ and $v$
   - Retain all other edges
3. After contraction:
   - Parallel edges may exist
   - Self-loops do not exist

### 1.4 Successful Min-Cut Run

**Step-by-step definition:**

1. A successful min-cut run:
   - Refers to success in the outcome of an algorithm
2. The algorithm:
   - Correctly identifies the minimum cut of the graph
3. The resulting cut corresponds to the true minimum cut

### 1.5 Unsuccessful Min-Cut Run

**Step-by-step definition:**

1. An unsuccessful min-cut run:
   - Refers to an iteration of a min-cut algorithm
2. The algorithm:
   - Fails to correctly identify the minimum cut
3. This occurs due to:
   - Unfavorable contraction choices during execution

### 1.6 Max-Flow Min-Cut Theorem

**Theorem statement (as given):**

> "In a flow network, the maximum amount of flow passing from the source to the sink is equal to the total weight of the edges in a minimum cut."

**Step-by-step definitions:**

1. **Capacity of a cut**:
   - Sum of capacities of edges oriented from vertex $\in X$ to vertex $\in Y$

2. **Minimum cut**:
   - The cut with the smallest possible capacity
3. **Minimum cut capacity**:
   - Capacity of the minimum cut
4. **Maximum flow**:
   - Largest possible flow from source $S$ to sink $T$

# 2. Deterministic Min-Cut Algorithm

## 2.1 Stoer–Wagner Minimum Cut Algorithm

**Theorem-based reasoning (step-by-step):**

1. Let $s$ and $t$ be two vertices of graph $G$
2. Let $G/\{s,t\}$ be the graph obtained by merging $s$ and $t$
3. A minimum cut of $G$ is:
   - The smaller of:
     - A minimum $s$-$t$ cut of $G$
     - A minimum cut of $G/\{s,t\}$
4. Two cases:
   - Case 1: A minimum cut separates $s$ and $t$
     - Then minimum $s$-$t$ cut is a minimum cut
   - Case 2: No minimum cut separates $s$ and $t$
     - Then minimum cut of $G/\{s,t\}$ is the answer

## 2.2 Deterministic Min-Cut Algorithm – Pseudocode

**Algorithm 1:** `MinimumCutPhase(G, a)`

```
A ← {a}
while A \neq V do
    add to A the most tightly connected vertex
return the cut weight as the \cut of the phase"
```

**Algorithm 2:** `MinimumCut(G)`

```
while |V| \geq 1 do
    choose any a from V
    MinimumCutPhase(G, a)
    if the cut-of-the-phase is lighter than the current minimum cut then
        store the cut-of-the-phase as the current minimum cut
    shrink G by merging the two vertices added last
return the minimum cut
```

# 3. Randomized Min-Cut Algorithm

## 3.1 Why Randomized Algorithm?

**Step-by-step points from lecture:**

1. Randomized algorithms:
   - Provide probabilistic guarantee of success
2. They:
   - Provide a more accurate estimate with fewer iterations
3. Additional listed properties:
   - Efficiency
   - Parallelization
   - Approximation guarantees
   - Avoidance of worst-case instances
   - Heuristic nature
   - Robustness

## 3.2 Karger's Randomized Algorithm

**Algorithmic idea (as shown):**

1. Randomly pick an edge
2. Remove it
3. Fuse its two endpoints
4. Repeat contraction until only two vertices remain
5. The remaining edges form a cut
6. The output cut may or may not be the minimum cut

## 3.3 Randomized Min-Cut Algorithm – Pseudocode

**Algorithm 3:** `RECURSIVE-RANDOMIZED-MIN-CUT(G, α)`

```
Input: undirected multigraph G with n vertices, integer alpha > 0
Output: a cut C of G
if n <= alpha^3 then
    C <- a min-cut of G found using brute force (exhaustive search)
else
    for i <- 1 to alpha do
        G' <- multigraph obtained by applying n - ceil(n / sqrt(alpha)) random contraction s
        C' <- RECURSIVE-RANDOMIZED-MIN-CUT(G', alpha)
        if i = 1 or |C'| < |C| then
            C <- C'
return C
```

### 3.4 Comparison: Deterministic vs Randomized Min-Cut Algorithm

**Decision basis:**

- Any specific problem is responsible for deciding the approach

**Deterministic Min-Cut:**

1. Always guarantees exact minimum cut
2. May have higher time complexity for large graphs
3. Stoer–Wagner time complexity:

$$O(V \cdot E + V^2 \log V)$$

**Randomized Min-Cut:**

1. Provides approximate minimum cut with high probability
2. Karger's algorithm time complexity:

$$O(V^2)$$

### 3.5 Theorem for Min-Cut Set

**Theorem (as stated):**

The algorithm outputs a min-cut set with probability at least $\dfrac{2}{n(n-1)}$

### 3.6 Python Simulation

**Lecture instruction:**

1. Open Campuswire post regarding Lecture 10
2. Download the provided `.ipynb` file
3. Use it for simulation of the randomized min-cut algorithm

### 3.7 Class Participation

- Participation linked to:
  - Campuswire discussion
  - Python simulation activity

**End of Lecture Scribe**